1. Proses instalasi minikube

```
Command Prompt

Microsoft Windows [Version 10.0.26200.8246]
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C:\Users\Hafizh>winget install Kubernetes.minikube
The `msstore` source requires that you view the following agreements before using.
Terms of Transaction: https://aka.ms/microsoft-store-terms-of-transaction
The source requires the current machine's 2-letter geographic region to be sent to the backend service to function prope
rly (ex. "US").

Do you agree to all the source agreements terms?
[Y] Yes  [N] No: y
Found Kubernetes - Minikube - A Local Kubernetes Development Environment [Kubernetes.minikube] Version 1.38.1
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://github.com/kubernetes/minikube/releases/download/v1.38.1/minikube-installer.exe
  48.5 MB / 48.5 MB
Successfully verified installer hash
Starting package install...
Successfully installed

C:\Users\Hafizh>
```

2. Jalankan minikube dan cek node

```
Command Prompt

Microsoft Windows [Version 10.0.26200.8246]
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C:\Users\Hafizh>minikube version
minikube version: v1.38.1
commit: c93a4cb9311efc66b90d33ea03f75f2c4120e9b0

C:\Users\Hafizh>minikube start --driver=docker
* minikube v1.38.1 on Microsoft Windows 11 Home Single Language 25H2
* Using the docker driver based on user configuration
! Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.
* Using Docker Desktop driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Downloading Kubernetes v1.35.1 preload ...
    > preloaded-images-k8s-v18-v1...:  272.45 MiB / 272.45 MiB  100.00% 6.75 Mi
    > gcr.io/k8s-minikube/kicbase...:  519.58 MiB / 519.58 MiB  100.00% 5.83 Mi
* Creating docker container (CPUs=2, Memory=4000MB) ...
* Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
* Configuring bridge CNI (Container Networking Interface) ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: storage-provisioner, default-storageclass
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default

C:\Users\Hafizh>kubectl get nodes
NAME       STATUS   ROLES           AGE     VERSION
minikube   Ready    control-plane   2m57s   v1.35.1

C:\Users\Hafizh>
```
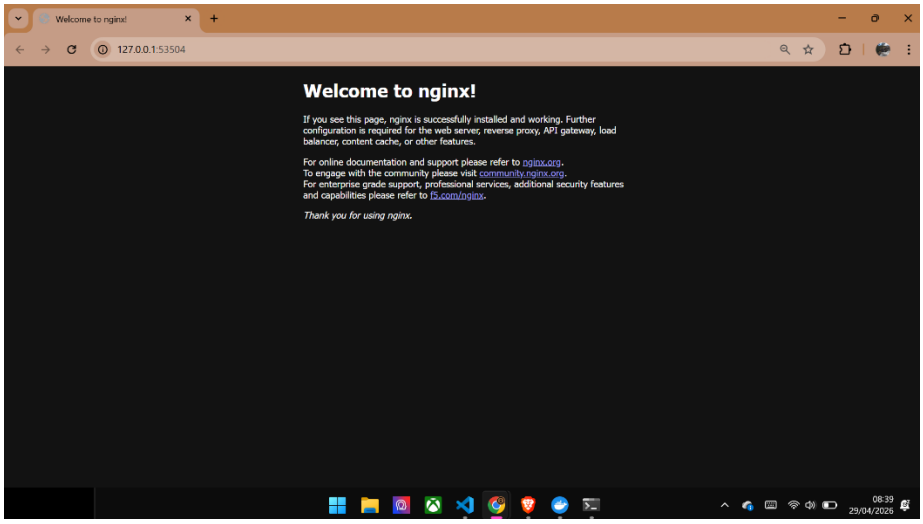

3. Set up folder project , apply file , cek pods dan membuka aplikasi

```
Command Prompt

C:\Users\Hafizh>mkdir kubernetes

C:\Users\Hafizh>cd kubernetes

C:\Users\Hafizh\kubernetes>code .

C:\Users\Hafizh\kubernetes>kubectl apply -f deployment.yaml
deployment.apps/web-nginx created

C:\Users\Hafizh\kubernetes>kubectl apply -f service.yaml
service/web-nginx-service created

C:\Users\Hafizh\kubernetes>kubectl get all
NAME                              READY   STATUS              RESTARTS   AGE
pod/web-nginx-748d764f7-95snp     0/1     ContainerCreating   0          24s
pod/web-nginx-748d764f7-h427h     0/1     ContainerCreating   0          24s
pod/web-nginx-748d764f7-mj2jf     0/1     ContainerCreating   0          24s

NAME                          TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
service/kubernetes            ClusterIP   10.96.0.1      <none>        443/TCP        9m59s
service/web-nginx-service     NodePort    10.108.91.17   <none>        80:30524/TCP   15s

NAME                          READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/web-nginx     0/3     3            0           25s

NAME                                     DESIRED   CURRENT   READY   AGE
replicaset.apps/web-nginx-748d764f7      3         3         0       25s

C:\Users\Hafizh\kubernetes>
```

```
C:\Users\Hafizh\kubernetes>kubectl get pods
NAME                          READY   STATUS    RESTARTS   AGE
web-nginx-748d764f7-95snp     1/1     Running   0          58s
web-nginx-748d764f7-h427h     1/1     Running   0          58s
web-nginx-748d764f7-mj2jf     1/1     Running   0          58s

C:\Users\Hafizh\kubernetes>kubectl get pods -w
NAME                          READY   STATUS    RESTARTS   AGE
web-nginx-748d764f7-95snp     1/1     Running   0          65s
web-nginx-748d764f7-h427h     1/1     Running   0          65s
web-nginx-748d764f7-mj2jf     1/1     Running   0          65s
```

```
C:\Users\Hafizh>minikube service web-nginx-service
|-----------|-----------------|-------------|---------------------------|
| NAMESPACE |      NAME       | TARGET PORT |           URL             |
|-----------|-----------------|-------------|---------------------------|
| default   | web-nginx-service | 80          | http://192.168.49.2:30524 |
|-----------|-----------------|-------------|---------------------------|
* Starting tunnel for service web-nginx-service.
|-----------|-----------------|-------------|---------------------------|
| NAMESPACE |      NAME       | TARGET PORT |           URL             |
|-----------|-----------------|-------------|---------------------------|
| default   | web-nginx-service |             | http://127.0.0.1:53504    |
|-----------|-----------------|-------------|---------------------------|
* Opening service default/web-nginx-service in default browser...
! Because you are using a Docker driver on windows, the terminal needs to be open to run it.
```

4. Hasil run aplikasi



5. Lihat pod live di terminal 1

```
Command Prompt - minikube    Windows PowerShell

Windows PowerShell
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PS C:\Users\Hafizh> kubectl get pods -w
NAME                          READY   STATUS    RESTARTS   AGE
web-nginx-748d764f7-95snp     1/1     Running   0          3m
web-nginx-748d764f7-h427h     1/1     Running   0          3m
web-nginx-748d764f7-mj2jf     1/1     Running   0          3m
web-nginx-748d764f7-brcgf     0/1     Pending   0          0s
web-nginx-748d764f7-brcgf     0/1     Pending   0          0s
web-nginx-748d764f7-48xfg     0/1     Pending   0          0s
web-nginx-748d764f7-48xfg     0/1     Pending   0          0s
web-nginx-748d764f7-brcgf     0/1     ContainerCreating   0          1s
web-nginx-748d764f7-48xfg     0/1     ContainerCreating   0          1s
web-nginx-748d764f7-brcgf     0/1     Terminating         0          8s
web-nginx-748d764f7-48xfg     0/1     Terminating         0          8s
web-nginx-748d764f7-brcgf     1/1     Terminating         0          9s
web-nginx-748d764f7-brcgf     0/1     Completed           0          11s
web-nginx-748d764f7-brcgf     0/1     Completed           0          12s
web-nginx-748d764f7-brcgf     0/1     Completed           0          12s
web-nginx-748d764f7-brcgf     0/1     Completed           0          12s
web-nginx-748d764f7-48xfg     1/1     Terminating         0          14s
web-nginx-748d764f7-48xfg     0/1     Completed           0          15s
web-nginx-748d764f7-48xfg     0/1     Completed           0          16s
web-nginx-748d764f7-48xfg     0/1     Completed           0          16s
web-nginx-748d764f7-48xfg     0/1     Completed           0          16s
```

6. Menghapus project

```
Command Prompt - miniku    Windows PowerShell    Windows PowerShell    Windows PowerShell

Windows PowerShell
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PS C:\Users\Hafizh> kubectl scale deployment web-nginx --replicas=5
deployment.apps/web-nginx scaled
PS C:\Users\Hafizh> kubectl scale deployment web-nginx --replicas=3
deployment.apps/web-nginx scaled
PS C:\Users\Hafizh>
```

7. Menambahkan Worker Node

```
PS C:\Users\Hafizh> minikube node add
😄  Adding node m02 to cluster minikube as [worker]
❗  Cluster was created without any CNI, adding a node to it might cause broken networking.
👍  Starting "minikube-m02" worker node in "minikube" cluster
🚜  Pulling base image v0.0.50 ...
🔥  Creating docker container (CPUs=2, Memory=2200MB) ...
❗  Failing to connect to https://registry.k8s.io/ from inside the minikube container
💡  To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networki
ng/proxy/
🐳  Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔎  Verifying Kubernetes components...
🏄  Successfully added m02 to minikube!
PS C:\Users\Hafizh> kubectl get nodes
NAME           STATUS     ROLES           AGE   VERSION
minikube       Ready      control-plane   18m   v1.35.1
minikube-m02   NotReady   <none>          69s   v1.35.1
PS C:\Users\Hafizh> minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

minikube-m02
type: Worker
host: Running
kubelet: Running

PS C:\Users\Hafizh>
```